Unit: Unit 3: Functions
Topic: Definition of Relation and Function, Domain, Range, and Notations

Name: ______________
Date: __________

Quiz (Carolina Reaper)

Q1. A trucking company has a relation between the number of packages delivered and the total weight of the packages. If the relation is $\{(1,10),(2,15),(3,10),(4,20)\}$, is this relation a function?
(A) Yes
(B) No
(C) Insufficient data
(D) Maybe
(E) Not applicable

Q2. The time it takes for a package to be delivered from a warehouse to a customer's doorstep is given by the function $f(x) = 2x + 5$, where x is the distance in miles. What is the delivery time for a package traveling 8 miles?
(A) 15 minutes
(B) 20 minutes
(C) 21 minutes
(D) 25 minutes
(E) 30 minutes

Q3. A shipping company has a graph showing the relationship between the number of packages shipped and the cost of shipping. If the graph passes the vertical line test, what can be concluded about the relationship?
(A) It is not a function
(B) It is a function
(C) It is a relation
(D) It is a table
(E) It is a mapping

Q4. The domain of the function $f(x) = \frac{1}{x}$ is $\{x \mid x \neq 0\}$. What is the range of this function?
(A) $\{y \mid y \neq 0\}$
(B) $\{y \mid y > 0\}$
(C) $\{y \mid y < 0\}$
(D) $\{y \mid y = 0\}$
(E) $\{y \mid y \geq 0\}$

Q5. A logistics company uses the function $f(x) = 3x - 2$ to determine the cost of shipping x packages. What is the cost of shipping 5 packages?
(A) 13
(B) 15
(C) 17
(D) 19
(E) 20

Q6. The range of the function $f(x) = x^2$ is $\{y \mid y \geq 0\}$. What is the domain of this function?
(A) $\{x \mid x \geq 0\}$
(B) $\{x \mid x > 0\}$
(C) $\{x \mid x < 0\}$
(D) $\{x \mid x = 0\}$
(E) $\{x \mid x \in \mathbb{R}\}$

Q7. A transportation company uses the relation $\{(1,2),(2,3),(3,4),(4,2)\}$ to determine the number of trucks needed to transport x packages. Is this relation a function?
(A) Yes
(B) No
(C) Insufficient data
(D) Maybe
(E) Not applicable

Q8. The function $f(x) = 2x + 1$ represents the time it takes for a package to be delivered from a warehouse to a customer's doorstep. What is the value of $f(3)$?
(A) 6
(B) 7
(C) 8
(D) 9
(E) 10

Open Ended Questions (FRQ)

Q9. A shipping company uses the function $f(x) = x^2 + 2x + 1$ to determine the cost of shipping x packages. Find the domain and range of this function and evaluate $f(2)$.

Q10. A logistics company uses the relation $\{(1,3),(2,5),(3,7),(4,5)\}$ to determine the number of trucks needed to transport x packages. Determine whether this relation is a function and find the domain and range if it is a function.

Chef's Tips

- Chef's Tips: Make sure to read each question carefully and choose the correct answer.
- Chef's Tips: Use the vertical line test to determine if a relation is a function.
- Chef's Tips: Evaluate functions using proper function notation.